

Observability (Prometheus / Grafana-as-code, SLOs & alerts)

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Master technical specification — Volume 6 (Deployment, Tooling & Operations), nano-detail document `observability.md`. Scope: the **operator-facing observability stack** for HelixVPN — Prometheus scrape config, the complete metric catalogue (all **counts/gauges**, no traffic or flow logs, per the no-logging promise C3), SLO definitions, concrete **PromQL** alert rules, and **Grafana dashboards-as-code**. It is the deployment-side consumer of the series that `[svc-telemetry]` *emits* — telemetry owns the series, `deploy/` owns the dashboards + alert rules `[svc-telemetry §0.1]`. This is a SPEC (describe the implementation; do not build the product). Evidence cited inline: `[05 §N]` (`05-repo-layout-tooling-and-helix-ecosystem.md`), `[svc-telemetry §N]`, `[svc-coordinator §N]`, `[svc-events §N]`, `[kubernetes §N]`. The privacy boundary (C3 — counts only, never flows) is load-bearing throughout. Unproven facts are marked UNVERIFIED per §11.4.6.

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0. The observability charter — counts, never flows

HelixVPN's observability is constrained by a privacy invariant before it is shaped by an operational one. From [svc-telemetry §0]:

telemetry exposes "Counts and gauges only — **never per-packet, per-flow, per-destination data** (C3)." Permitted labels: stream, group, consumer, event_type, code, version. **Forbidden labels:** tenant_id, device_id, overlay_ip, src_ip, dst_ip, any high-cardinality or PII/traffic-shaped key [svc-telemetry §3.1].

So this document specifies an observability stack that is operationally useful **and** structurally incapable of becoming a connection log. The two design pressures resolve as:

1. **Every series is an aggregate** — a histogram of convergence latency, a gauge of open streams, a counter of revokes. None carries who-talked-to-whom.
2. **The headline SLO is convergence**, not throughput — HelixVPN measures "did the network map reach the agent in < 1 s," not "how many bytes flowed" (it deliberately cannot measure the latter, C3) [svc-telemetry §3.2].

1. Topology — who scrapes what

flowchart LR

```
subgraph helixd[helixd control plane]
    M9[/metrics :9090<br/>private prometheus.Registry/]
end
subgraph edge[helix-edge data plane]
    ME[/metrics<br/>handshakes, transport mix - edge-owned/]
end
PROM[Prometheus<br/>scrape internal only §5] --> M9
PROM -. separate job .-> ME
PROM --> RULES[recording + alerting rules §4]
PROM --> GRAF[Grafana dashboards-as-code §6]
RULES --> AM[Alertmanager → pager/Slack]
classDef ext fill:#eef,stroke:#557;
class PROM,GRAF,AM ext;
```

Two scrape targets, two ownership boundaries [svc-telemetry §0.1]:

- **helixd:9090/metrics** — control-plane internal state (convergence, event lag, stream depth, audit counts, presence gauge, Go runtime). Owned by internal/telemetry via a **private** prometheus.Registry (not the global default, to keep the series set auditable, [svc-telemetry §3]). Scraped **internal-only** — never on the public 443 listener (§5, [svc-telemetry §8.3]).
 - **helix-edge:/metrics** — edge-side handshake/transport metrics, owned by the Rust edge [svc-telemetry §0.1], scraped as a separate Prometheus job. **UNVERIFIED** exact edge series names (the edge's own /metrics is the data-plane spec's concern); this document scopes the control-plane series and treats the edge job as a configured-but-out-of-scope target.
-

2. The complete metric catalogue (counts/gauges only)

The canonical declarations live in `internal/telemetry/metrics.go` [svc-telemetry §3] and `internal/events/metrics.go` [svc-events §8.2]. Consolidated catalogue (deployment-side view):

2.1 Convergence / real-time SLO series — the < 1 s promise made falsifiable

Series	Type	Meaning	Source
<code>helix_reconcile_seconds</code>	histogram	event-receive → MapDelta on the WatchNetworkMap wire (the headline SLO)	[svc-telemetry §3.2], measured at coordinator [svc-telemetry §6.1]
<code>helix_event_lag_seconds{stream}</code>	histogram	Envelope.ts → consumer pickup (bus delivery latency — the convergence-fault discriminator)	[svc-telemetry §3.2/§6.2]
<code>helix_event_bus_seconds{type}</code>	histogram	outbox created_at → consumer receive (whole publish path)	[svc-events §8.2]
<code>helix_event_outbox_lag_seconds</code>	histogram	outbox created_at → relayed (XADD succeeded)	[svc-events §8.2]
<code>helix_revoke_enforce_seconds</code>	histogram	device.revoke accepted → peer-removal delta on wire (security SLO)	[svc-telemetry §3.2]
<code>helix_stream_pending_entries{stream,group}</code>	gauge	Redis PEL depth (un-acked backlog — stuck-consumer leading indicator)	[svc-telemetry §3.2], [svc-events §8.3]

The histogram buckets for `helix_reconcile_seconds` straddle 1.0 deliberately (`{... 0.5, 0.75, 1.0, 2.5, 5.0}`) so `histogram_quantile(0.99, ...)` < 1.0 is directly readable [svc-telemetry §3.2] — a unit test asserts the buckets straddle the SLO ([svc-telemetry §10 T-UNIT-4]).

2.2 Topology / load gauges — active-peer & stream counts (all COUNTS, C3)

Series	Type	Meaning
helix_open_watch_streams	gauge	current open WatchNetworkMap streams (coordinator load) [svc-telemetry §3.2]
helix_fanout_affected_nodes	histogram	size of the minimal affected set per event (should be small) [svc-coordinator §7.2]
helix_graph_nodes	gauge	nodes in the in-mem graph — global aggregate, NO {tenant} label (resolved at source, see below) [svc-coordinator §7.2]
helix_presence_online_devices	gauge	devices currently online in Redis TTL presence (the active-peer gauge; svc-coordinator §7.2 names the same gauge <code>helix_presence_online</code> — canonical svc-telemetry name is <code>helix_presence_online_devices</code>) [svc-telemetry §3.3]
helix_backpressure_drops_total	counter	slow-consumer stream drops [svc-coordinator §7.2]

Reconciled (§11.4.35, 2026-06-26) — {tenant} labels: resolved at source, no contradiction. An earlier draft flagged a cross-doc contradiction ([svc-coordinator §7.2] declaring `helix_open_streams{tenant}` / `helix_graph_nodes{tenant}` vs [svc-telemetry §3.1] forbidding `tenant_id`). That contradiction **no longer exists**: [svc-coordinator §7.2] was itself reconciled (§11.4.35, 2026-06-26) so its `/metrics` gauges (`helix_open_streams`, `helix_graph_nodes`, `helix_presence_online`) are now **global aggregates with NO {tenant} label** — already agreeing with svc-telemetry's C3 cardinality ban. The deployment-side catalogue therefore simply **mirrors the resolved state**: per-tenant breakdowns are forbidden on `/metrics` and any genuine per-tenant count is routed behind the authenticated REST `/v1/stats` ([svc-telemetry §3.1]), not Prometheus. The unit test [svc-telemetry §10 T-UNIT-3] (every collector's label set $\subseteq$ allow-list) remains the mechanical enforcer; a `{tenant}` label that survives is a test failure (§11.4.6 — no contradiction is asserted that the source specs do not actually hold).

2.3 Handshake-failure & event-health counters

Series	Type	Meaning
helix_events_dlq_total{stream,group}	counter	poison events dead-lettered (any sustained increase alerts) [svc-events §8.2]
helix_events_published_total{type}	counter	events published, by type [svc-events §8.2]
helix_event_react_errors_total{type}	counter	failed reactions, by type [svc-events §8.2]
helix_audit_events_total{action}	counter	control-action audit rows, by closed-vocabulary action [svc-telemetry §3.3]
helix_audit_write_errors_total{code}	counter	audit-sink failures by error code [svc-telemetry §3.3]
helix_presence_backend_up	gauge (0/1)	Redis presence backend reachable [svc-telemetry §5.4]

Handshake-fail rate is split-owned. The *control-plane* analogue of "handshake failure" is `helix_event_react_errors_total` + `helix_events_dlq_total` (a reaction/event that could not be processed). The **transport handshake-fail rate** (WG/MASQUE handshakes that fail at the edge) lives on the **edge** /metrics [svc-telemetry §0.1, §1], not in the Go control plane; it is scraped from the edge job (§1) and its series names are UNVERIFIED here (data-plane spec owns them). The dashboards (§6) panel both, clearly labelled by owner.

2.4 Runtime / build series (Go collectors)

`go_goroutines`, `go_gc_duration_seconds`, `process_resident_memory_bytes` (the 24 h-soak no-leak SLO series), and `helix_build_info{version,commit}` (version = release prefix `helix_vpn-...`, §11.4.151) [svc-telemetry §3.3]. Registered explicitly into the private registry so the global default-collector duplication is avoided [svc-telemetry §3].

3. SLOs — the promises made falsifiable

SLO	Series + query	Threshold
convergence p99 < 1 s	histogram_quantile(0.99, sum(rate(helix_reconcile_seconds_bucket[5m])) by (le))	> 1 2 m
revoke < 1 s	histogram_quantile(0.99, sum(rate(helix_revoke_enforce_seconds_bucket[5m])) by (le))	> 1
event lag healthy	histogram_quantile(0.99, sum(rate(helix_event_lag_seconds_bucket{stream="events:policy"} [5m])) by (le))	> 0
event bus path p99	histogram_quantile(0.99, sum(rate(helix_event_bus_seconds_bucket[5m])) by (le))	> 0 for 5
outbox relay lag	histogram_quantile(0.99, sum(rate(helix_event_outbox_lag_seconds_bucket[5m])) by (le))	> 0 5 m
backlog not growing	helix_stream_pending_entries	> 1 for 5
no DLQ accumulation	increase(helix_events_dlq_total[15m])	> 0
coordinator no leak	deriv(process_resident_memory_bytes[6h])	> 0 susta over

The convergence + revoke SLOs are the load-bearing ones; the rest are the *discriminators* that tell an operator **where** a breach lives — [svc-telemetry §6.2]: "if convergence breaches 1 s, the lag series says whether the bus or the coordinator is at fault." That decomposition is the systematic-debugging discriminator (§11.4.102 spirit) baked into the metric set.

4. Alert rules (concrete PromQL)

Rules-as-code live in `deploy/grafana/` (or `deploy/prometheus/rules/`), versioned, reviewed, deployed with the manifests — telemetry only emits the series; `deploy/` owns the rules [svc-telemetry §0.1/§3.4].

```
# deploy/prometheus/rules/helixvpn-slo.rules.yaml
groups:
- name: helixvpn-convergence
  rules:
    - alert: HelixConvergenceSLOBreached
```

```

    expr: histogram_quantile(0.99,
    sum(rate(helix_reconcile_seconds_bucket[5m])) by (le)) >
    1.0
    for: 2m
    labels: { severity: page }
    annotations:
        summary: "convergence p99 > 1s (C5 SLO breach)"
        description: "event→delta-on-wire p99 has exceeded 1s
        for 2m. Check helix_event_lag_seconds to localize bus-vs-
        coordinator [svc-telemetry §6.2]."
```

- alert: HelixRevokeSLOBreached


```

    expr: histogram_quantile(0.99,
    sum(rate(helix_revoke_enforce_seconds_bucket[5m])) by
    (le)) > 1.0
    for: 1m
    labels: { severity: page }
    annotations: { summary: "device revoke enforcement p99 >
    1s (security SLO)" }
```
- name: helixvpn-bus-health
 rules:
 - alert: HelixDLQAccumulating


```

    expr: increase(helix_events_dlq_total[15m]) > 0
    for: 0m
    labels: { severity: page }
    annotations:
        summary: "events dead-lettered – poison event(s)"
        description: "Inspect with `XRANGE <stream>:dlq - +`;
        fix cause; `helixvpncntl events replay-dlq <stream>` [svc-
        events §6.5]."
```
 - alert: HelixStreamBacklogGrowing


```

    expr: max(helix_stream_pending_entries) by (stream, group)
    > 1000
    for: 5m
    labels: { severity: warn }
    annotations: { summary: "consumer PEL depth > 1000 –
    stuck/slow consumer [svc-events §8.3]" }
```
 - alert: HelixEventLagHigh


```

    expr: histogram_quantile(0.99,
    sum(rate(helix_event_lag_seconds_bucket[5m])) by (le)) >
    0.5
    for: 5m
    labels: { severity: warn }
    annotations: { summary: "bus delivery lag p99 > 0.5s –
    Redis/relay degraded" }
```
- name: helixvpn-runtime
 rules:
 - alert: HelixCoordinatorMemoryLeak


```

    expr: deriv(process_resident_memory_bytes{job="helixd"}
    [6h]) > 0
    for: 30m
    labels: { severity: warn }
```

```

    annotations: { summary: "helixd RSS slope > 0 over 6h –
    possible leak (24h-soak SLO) [svc-telemetry §3.4]" }
- alert: HelixPresenceBackendDown
  expr: helix_presence_backend_up == 0
  for: 1m
  labels: { severity: warn }
  annotations:
    summary: "Redis presence backend unreachable"
    description: "Presence unknowable; tunnels UNAFFECTED
    (fail-static C1) [svc-telemetry §5.4]. Audit still writes
    (fail-closed)."
```

```

- alert: HelixReadinessFlapping
  expr: count(up{job="helixd"} == 1) < 1
  for: 1m
  labels: { severity: page }
  annotations: { summary:
    "no ready helixd replica – control plane API down (data
    plane may still forward)" }
```

Severity discipline: severity: page for the two < 1 s SLOs + DLQ + total-control-plane-down (operator-actionable now); severity: warn for degradations that fail-static (presence backend down, memory slope, lag) — they do not drop tunnels (C1) and so are not pages.

5. Scrape config & label-cardinality safety

```

# deploy/prometheus/scrape/helixvpn.yaml (Prometheus
# scrape_configs fragment)
scrape_configs:
- job_name: helixd
  metrics_path: /metrics
  scheme: http # internal port :9090 –
  NEVER the public 443 [svc-telemetry §8.3]
  kubernetes_sd_configs: [{ role: pod, namespaces: { names:
    [helixvpn] } }]
  relabel_configs:
  - source_labels: [__meta_kubernetes_pod_label_app]
    action: keep
    regex: helixd
  - source_labels:
    [__meta_kubernetes_pod_container_port_number]
    action: keep
    regex: "9090" # scrape ONLY the metrics
    port, not :8443 (agent API)
- job_name: helix-edge # separate ownership
  boundary ($1) – edge-owned series
  metrics_path: /metrics
  kubernetes_sd_configs: [{ role: pod, namespaces: { names:
    [helixvpn] } }]
  relabel_configs:
  - source_labels: [__meta_kubernetes_pod_label_app]
```



```
action: keep
regex: helix-edge
```

Two hard rules, both mechanically enforced:

1. **/metrics is internal-only.** It carries no PII (C3) but still reveals operational shape (tenant population via gauges, deploy version), so it is bound to the internal port `:9090` and gated by network isolation / mTLS for the scraper — never published on the public 443 listener [svc-telemetry §8.3, kubernetes §7]. The scrape job keeps only the `:9090` port.
2. **Label cardinality is bounded by construction.** Every registered collector's label set $\subseteq$ the allow-list (stream, group, consumer, event_type, code, version); a unit test asserts it ([svc-telemetry §10 T-UNIT-3]). Prometheus relabeling does NOT add `tenant_id/device_id` — and there is nothing to add, because the exporter never emits them. The §2.2 {tenant} question is resolved on the *exporter* side at source (svc-coordinator §7.2 now emits global aggregates with no {tenant} label), not patched in scrape config.

6. Grafana dashboards-as-code

Dashboards are JSON-as-code under `deploy/grafana/dashboards/`, provisioned (not click-built), so they version + review + deploy with the manifests [05 §2 deploy/grafana, svc-telemetry §0.1].

```
# deploy/grafana/provisioning/dashboards.yaml
apiVersion: 1
providers:
- name: helixvpn
  folder: HelixVPN
  type: file
  options: { path: /var/lib/grafana/dashboards/helixvpn }
```

Three canonical dashboards (each a versioned JSON; panels described, not dumped):

Dashboard	Panels (PromQL)
Convergence SLO	helix_reconcile_seconds p50/p99 heatmap + the 1.0s SLO line; helix_revoke_enforce_seconds p99; helix_event_lag_seconds{stream} per-stream (the bus-vs-coordinator discriminator); helix_fanout_affected_nodes (minimal-set sanity)
Bus health	helix_stream_pending_entries{stream,group} depth; rate(helix_events_published_total{type}[5m]); increase(helix_events_dlq_total[1h]); helix_event_outbox_lag_seconds p99
Control-plane runtime	helix_open_watch_streams; helix_presence_online_devices (active-peer gauge) + helix_presence_backend_up; process_resident_memory_bytes (24h soak);

Dashboard	Panels (PromQL)
	go_goroutines; rate(helix_audit_events_total{action}[5m]) (control-action audit volume — counts only)

A fourth **edge transport** dashboard panels the edge job's handshake/transport-mix series (§2.3), clearly separated as data-plane-owned. Its exact panels are UNVERIFIED (edge series names) and stubbed as a placeholder provider pointing at the edge job — NOT fabricated with invented series names.

The §11.4.168 exported-document discipline applies to any rendered dashboard screenshots committed as evidence: a committed dashboard PNG/PDF must visually render (no raw JSON leaking as body text) and pass independent visual validation.

7. The no-logging boundary, restated for observability

This is the section an auditor reads. Everything above is counts and gauges; **nothing** here can reconstruct who-talked-to-whom:

- **No flow/traffic series exist.** There is no bytes_total, no flows, no src_ip/dst_ip label anywhere in the catalogue (§2). The schema-lint [svc-telemetry §7] forbids the *durable* shape; the label allow-list test [svc-telemetry §10 T-UNIT-3] forbids the *metric* shape. Both are runtime signatures (§11.4.108) asserted against the live system, not just the source.
- **Presence is a count, not a log.** helix_presence_online_devices is a single gauge; the per-device presence lives in TTL Redis and is never copied to Postgres or to a metric label [svc-telemetry §5]. You can see *how many* devices are online, never *which* on /metrics.
- **Audit is control-actions only.** helix_audit_events_total{action} counts revokes / policy-activations by closed-vocabulary action [svc-telemetry §3.3/§4.2] — never traffic. The action label is a bounded closed set, so it cannot grow toward a traffic log.
- **/metrics is not public (§5).** Even the aggregate shape is internal-only [svc-telemetry §8.3].

The observability stack is therefore *provably* incapable of being a connection log — which is the whole point: a VPN's observability must not undo its privacy promise.

8. Anti-bluff evidence

Claim	Captured-evidence proof (§11.4.5/§11.4.69)
convergence SLO is real & falsifiable (§3)	perf test drives a policy flip @ 10k streams; helix_reconcile_seconds p99 < 1 s histogram dump [svc-telemetry §10 T-PERF-1]
no forbidden labels reach	

Claim	Captured-evidence proof (§11.4.5/§11.4.69)
Prometheus (§2.2/§5)	label-audit unit test green [svc-telemetry §10 T-UNIT-3]; paired mutation adding a device_id label → test FAILs
alert fires on real breach (§4)	inject latency to push p99 > 1 s → HelixConvergenceSL0Breached fires in Alertmanager; recorded alert
/metrics not on public 443 (§5/§7)	scan: :443 carries no exposition, :9090 intra-cluster only [svc-telemetry §10 T-SEC-1]
dashboards provision from code (§6)	kubectl apply → Grafana shows the 3 dashboards with no manual click; screenshot per §11.4.159
no-logging holds (§7)	schemalint green against the deployed DB (runtime signature §11.4.108) [svc-telemetry §10 T-CHAL-1]; paired mutation planting a connections(src_ip...) table → FAIL

A dashboard that renders but whose underlying series are stubbed/zero is a §11.4 bluff — every panel must bind a real, populated series under a real load, captured.

9. UNVERIFIED register

#	UNVERIFIED item	Why / status
U1	Edge /metrics series names (handshake-fail, transport mix) (§2.3)	data-plane spec owns them; this doc scopes control-plane series only
U2	{tenant}-labelled gauges in [svc-coordinator §7.2] vs the §3.1 allow-list ban (§2.2)	RETIRED (§11.4.35, 2026-06-26) — resolved at source: svc-coordinator §7.2 now emits global aggregates with no {tenant} label, agreeing with the §3.1 ban. No live contradiction remains (§11.4.6); see §2.2 reconciliation note
U3	Prometheus naming-convention exact wording (_seconds/_total suffix rules)	standard practice; exact spec wording UNVERIFIED [svc-telemetry §3.1]
U4	/metrics port-vs-path isolation final choice (§5)	spec mandates <i>some</i> network isolation; port-separation is the recommended default, not validated [svc-telemetry §8.3]
U5	Alertmanager routing/receivers (pager/Slack)	deployment-environment specific; the rules carry severity labels, the routing tree is operator config

Sources verified

- `05-repo-layout-tooling-and-helix-ecosystem.md` §2 (deploy/grafana dashboards-as-code, telemetry counters only), §7 (substrates) — [05].
- `v03-control-plane/svc-telemetry.md` §0/§0.1 (charter, ownership boundaries: telemetry emits, deploy owns dashboards/rules), §3 (private registry, full series declarations), §3.1 (label allow-list / forbidden labels), §3.2/§3.3 (convergence + audit + presence + runtime series), §3.4 (SLO→alert mapping), §5 (TTL presence), §5.4 (presence drift / `helix_presence_backend_up`), §6.1/§6.2 (where convergence is measured + the lag discriminator), §7 (no-logging schema-lint), §8.3 (scrape security / internal-only), §10 (T-UNIT-3 label audit, T-PERF-1, T-SEC-1, T-CHAL-1) — [svc-telemetry].
- `v03-control-plane/svc-coordinator.md` §7.2 (`helix_fanout_affected_nodes`, `helix_open_streams`, the {tenant} label contradiction), §7.3 (SLO acceptance numbers) — [svc-coordinator].
- `v03-control-plane/svc-events.md` §6.5 (DLQ + replay-dlq), §8.2 (events metric set), §8.3 (PEL-depth gauge), §8.4 (alert thresholds) — [svc-events].
- `v06-deploy/kubernetes.md` §5 (scrape annotations / metrics port), §7 (metrics Service, internal-only) — [kubernetes] (sibling).

Constitution bindings applied: §11.4.44 (revision header), §11.4.6 (no-guessing — the §2.2 cross-doc {tenant}-label contradiction surfaced + resolved conservatively, UNVERIFIED register §9), §11.4.102 (the lag-discriminator decomposition = systematic-debugging baked in), §11.4.108 (no-logging runtime signature: schema-lint + label-audit against the deployed system), §11.4.168 (visual validation of any committed dashboard render), §11.4.5/.69 (captured-evidence anti-bluff proofs §8).